

Medicare Physician & Other Practitioners

Provider Utilization & Payment Analysis

Data Source: Centers for Medicare & Medicaid Services (CMS) | data.gov

Dataset: Medicare Physician & Other Practitioners - by Provider and Service (2023)

Tool: Microsoft Power BI (Power Query, DAX)

Records: 9.6 Million+ | Providers: 1.18 Million | HCPCS Codes: 6,405

1. Executive Summary

This project analyzes the CMS Medicare Physician & Other Practitioners dataset, which contains 9.7 million records representing every service billed to Medicare Part B by 1.18 million unique healthcare providers across 6,405 distinct procedure codes (HCPCS) in 2023. The analysis examines provider billing patterns, Medicare reimbursement efficiency, service utilization intensity, and geographic variation across all 50 U.S. states, the District of Columbia, and U.S. territories.

The goal is to uncover actionable insights into how Medicare spending is distributed across provider types, specialties, and geographies, and to identify where the largest gap exists between what providers charge and what Medicare actually pays.

2. Dataset Overview

Each row in the dataset represents a unique provider-service combination: one physician or practitioner (identified by NPI) performing one specific procedure (identified by HCPCS code). The dataset includes provider demographics, geographic information, service volume, and financial metrics.

2.1 Key Metrics & Definitions

Metric	Definition
Avg_Sbmtld_Chrg	Average amount the provider charged Medicare for the service
Avg_Mdcr_Alowd_Amt	Average amount Medicare deemed allowable for the service
Avg_Mdcr_Pymt_Amt	Average amount Medicare actually paid the provider
Avg_Mdcr_Stdzd_Amt	Standardized payment amount adjusted for geographic cost differences
Tot_Srvcs	Total number of services performed for the provider-HCPCS combination
Tot_Benes	Total unique Medicare beneficiaries who received the service

2.2 Calculated Measures (DAX)

Because the payment columns are pre-averaged at the provider-service level, simple averages across rows would be misleading. A provider who saw 12 patients would be weighed equally with one who saw 2,988. To address this, all financial aggregations use weighted averages:

Measure	Formula / Logic
Weighted Avg Medicare Payment	$\text{SUMPRODUCT}(\text{Avg_Mdcr_Pymt_Amt} \times \text{Tot_Srvcs}) / \text{SUM}(\text{Tot_Srvcs})$
Weighted Avg Submitted Charge	$\text{SUMPRODUCT}(\text{Avg_Sbmtld_Chrg} \times \text{Tot_Srvcs}) / \text{SUM}(\text{Tot_Srvcs})$
Change-to-Payment Ratio	Weighted Avg Submitted Charge / Weighted Avg Medicare Payment

Utilization Intensity	SUM(Tot_Srvcs) / SUM(Tot_Benes) average services per beneficiary
Payment Efficiency %	Weighted Avg Medicare Payment / Weighted Avg Submitted Charge

3. Key Findings

3.1 Provider Billing & Payment Analysis

The overall weighted average submitted charge across all providers is \$140.34, while Medicare pays only \$35.43 on average, a charge-to-payment ratio of 3.96. This means providers charge roughly four times what they receive from Medicare.

Ambulatory Surgical Centers command the highest weighted average Medicare payment at \$220.62, with a weighted average submitted charge of \$1,362.73. While individual ASC procedures can approach \$99,999, the weighted average is pulled down significantly because the vast majority of ASC services are routine, lower-cost outpatient procedures such as colonoscopies and cataract removals.

Radiation Therapy Centers have the highest weighted average submitted charge at \$1,704.61, reflecting the capital-intensive nature of radiation equipment and treatment delivery.

3.2 Charge-to-Payment Ratio

The charge-to-payment ratio reveals which specialties set prices furthest above Medicare reimbursement. Anesthesia-related specialties dominate the top of this ranking:

Provider Type	Ratio	Payment Efficiency
Anesthesiology Assistant	14.50	6.9%
Dental Anesthesiology	12.22	8.2%
Certified Registered Nurse Anesthetist	11.54	8.7%
Anesthesiology	10.38	9.6%
Radiation Therapy Center	8.81	11.4%

In contrast, Public Health or Welfare Agencies have the highest payment efficiency at 90.85% with a charge-to-payment ratio of just 1.10, indicating their charges are closely aligned with the Medicare fee schedule. Opioid Treatment Programs also show high efficiency at 83.07%.

3.3 Utilization Intensity

Utilization intensity (services per beneficiary) reveals how frequently patients within a specialty require repeated visits. The highest-intensity specialties are chronic disease and infusion-heavy providers:

Provider Type	Utilization	Total Services	Efficiency
Home Infusion Therapy Services	35.77	73,864	31%

Rheumatology	31	-	32%
Hematology-Oncology	22	-	-
Medical Oncology	22	-	-
Opioid Treatment Program	20	-	83%

A critical insight is that high utilization intensity does not correlate with high total service volume. Home Infusion Therapy has the highest intensity (35.77) but is among the lowest in total services at just 73,864, a small patient population requiring very frequent treatments. Clinical Laboratory presents the inverse: the highest total service volume at 380.4 million but a utilization intensity of only 2.57. This distinction highlights two fundamentally different models of Medicare spending wide and shallow (labs, office visits) versus narrow and deep (infusion therapy, oncology).

3.4 Service Volume Analysis

The top service by total volume is the provision of COVID-19 self-administered tests at 152.4 million services, far exceeding the second-ranked service (established patient office visits at 99 million). The total services across the dataset reach 2.64 billion.

At the state level, Florida leads with 287 million total services, followed by California (268M) and Texas (217M), broadly tracking Medicare-eligible population concentration. Within Florida, COVID-19 test provision is the largest contributor at 17.01 million services, with daptomycin injections second at 15.04 million.

At the provider type level, Clinical Laboratory accounts for 380.4 million total services — the highest of any provider type. Medicare Diabetes Preventive Program has the lowest at just 47 total services.

3.5 Geographic Analysis

American Samoa (AS) stands out as a significant outlier with the highest charge-to-payment ratio at 19.29 and the highest weighted average submitted charge, despite having the lowest total service volume at just 86 services. This extreme ratio likely reflects a very small number of providers with unusually high charge patterns. Puerto Rico (PR) has the lowest charge-to-payment ratio, suggesting provider charges there are most closely aligned with Medicare reimbursement.

Metropolitan areas show slightly higher weighted average submitted charges (\$141.24) compared to non-metro areas (\$133). However, the weighted average Medicare payment is nearly identical between metro (\$35.40) and non-metro (\$35.10) areas. This suggests that while providers in metro areas charge more, Medicare reimbursement is effectively standardized regardless of geographic setting, and the pricing gap is absorbed by the providers.

4. Methodology

4.1 Data Preparation

- Data sourced from CMS via data.gov (Medicare Physician & Other Practitioners — by Provider and Service, dated 2023-12-31)
- Loaded into Power BI via Power Query; unnecessary columns removed to optimize performance on 9.7M rows
- State/territory latitude and longitude lookup table merged via Power Query for accurate map geocoding
- Geographic classification (Metro vs. Non-Metro) derived from the RUCA codes in the dataset
- All financial measures calculated as weighted averages using Tot_Srvcs as the weight to account for volume differences across provider-service rows

4.2 Tools & Techniques

- Power BI Desktop: Data modeling, DAX measures, interactive dashboard design
- Power Query (M): Data cleaning, column transformations, table merges, and conditional columns
- DAX: Weighted average calculations, DIVIDE function for safe division, calculated measures for KPIs
- Python (pandas): Supplementary median calculations to overcome Power BI resource limitations on 9.7M rows

5. Dashboard Overview

The Power BI report consists of four interactive dashboard pages:

Provider Type Analysis: Top 20 provider types ranked by weighted average Medicare payment and submitted charge, with charge-to-payment ratios. Includes a bar chart of the highest charge-to-payment ratios and utilization intensity by provider type.

Service Volume Analysis: Top 15 HCPCS codes by total service amount, with KPI cards showing total services (2.64B), unique NPIs (1.18M), total records (9.7M), and distinct HCPCS codes (6,405).

Geographic Analysis: Interactive map visual showing total services, weighted average Medicare payment, and submitted charge by state. Includes state-level tables, a slicer for state filtering, charge-to-payment ratio by state, and Metro vs. Non-Metro comparison.

Utilization & Efficiency: Scatter plot of utilization intensity vs. weighted average Medicare payment (bubble size = total services), with KPI cards and bar charts for utilization intensity and payment efficiency by provider type.

6. Conclusions & Implications

This analysis reveals several structural patterns in Medicare Part B spending:

1. **The pricing gap is systemic.** Providers charge 4x what Medicare pays on average. Anesthesia-related specialties show the most extreme gaps (10–15x), suggesting these specialties set list prices with the expectation that Medicare will pay a small fraction.
2. **Volume and intensity are inversely related.** The highest-volume services (labs, office visits) have low per-beneficiary intensity, while the highest-intensity services (infusion therapy, oncology) serve small populations. These two models would require fundamentally different cost management strategies.
3. **COVID-19 testing dominates service volume.** At 152.4 million services, COVID-19 self-administered tests represent a significant portion of Medicare utilization and may skew aggregate metrics if not accounted for in analysis.
4. **Geographic variation exists in charges but not in payments.** Metro providers charge more than non-metro providers, but Medicare pays nearly the same amount in both settings. The standardized payment amount (Avg_Mdcr_Stdzd_Amt) effectively neutralizes geographic cost differences.
5. **Outlier territories require contextual interpretation.** American Samoa's extreme charge-to-payment ratio (19.29) and very low service volume (86 services) make it a statistical outlier. Insights from such small populations should be flagged rather than used for broad conclusions.